

# SpaceX Launch/Landing Analysis

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# Executive Summary

An analysis of SpaceX's success with their reusable Falcon 9 platform using applied data science to evaluate historical trends and other factors impact on future launch successes.

## Methodology:

- Data collection via SpaceX REST API and Wikipedia web scraping
- Data wrangling and labeling of landing outcomes
- Exploratory Data Analysis with visualization and SQL
- Interactive visual analytics with Folium and Plotly Dash
- Predictive analysis using four classification models tuned with GridSearchCV

## Key Results

- Launch success rate climbed from 0% (2010-2013) to ~85-90% by 2019-2020
- KSC LC-39A is the top-performing site: 10 successes, 76.9% success rate
- Payloads in the 2,000-5,500 kg range land most reliably
- Orbits ES-L1, GEO, HEO, and SSO show a 100% success rate
- Decision Tree classifier was the best model, with 87.5% cross-validated accuracy
- The ability to re-use the booster rockets provides a strong competitive advantage



# Introduction

## Project Background

SpaceX advertises Falcon 9 launches at roughly \$62 million, far below the \$165 million+ charged by competitors. Much of that savings comes from reusing the rocket's first stage. If the first stage's landing outcome can be predicted from public launch data, the cost of a launch can be estimated in advance — useful information for any company considering bidding against SpaceX.

## Questions to Answer

- How do payload mass, launch site, flight number, and orbit type affect first-stage landing success?
- Has the landing success rate improved over time?
- Which classification algorithm best predicts landing outcome for this dataset?



# Overall Methodology

- Three Data Sources were identified: SpaceX API, Wikipedia,
- Data collection: SpaceX REST API requests and Wikipedia web scraping (BeautifulSoup)
  - Filtering to Falcon 9 launches and imputing missing payload mass with the column mean
- Data wrangling: derived a binary landing Class label from the raw Outcome column
- Exploratory Data Analysis using Matplotlib/Seaborn visualizations and SQL queries
- Interactive visual analytics with a Folium map and a Plotly Dash web app
- Predictive analysis: standardized features, train/test split, and GridSearchCV (cv=10) tuning of
  - Logistic Regression, SVM, Decision Tree, and KNN classifiers

A decorative header consisting of a row of hexagons in various shades of blue, orange, and white. Below this row is another row of hexagons, some of which are outlined in white or orange.

# Data Collection

Data has been gathered from two main sources:

- SpaceX API – interacting with several endpoints SpaceX provides via their RESTful API, data was able to be retrieved directly from SpaceX
- Wikipedia – additional data was retrieved via web-scraping a publicly available Wikipedia page containing historic Falcon 9 launch data.

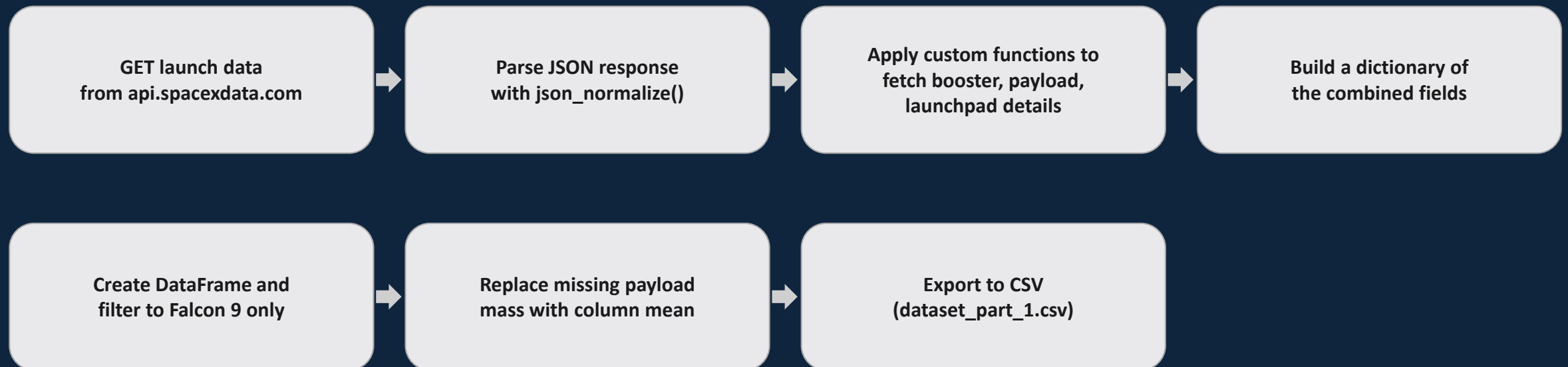


# Data Collection – SpaceX API (overview)

- The following RESTful APIs were used to retrieve JSON objects which were further processed in Python
- <https://api.spacexdata.com/v4/rockets/{rocketName}>
- <https://api.spacexdata.com/v4/launchpads/{launchpadName}>
- <https://api.spacexdata.com/v4/payloads/{mass}>
- <https://api.spacexdata.com/v4/cores/{core}>
- <https://api.spacexdata.com/v4/launches/past>

# Data Collection - SpaceX API (methodology)

Flowchart describes processing pipeline used to gather and cleans data retrieved via SpaceX API





# Data Collection - SpaceX API (results)

The final result was exported to dataset\_part\_1.csv, consisting of 90 records formatted as follows:

|    | A            | B         | C              | D                 | E        | F            | G           | H       | I        | J      | K     | L          | M     | N           | O      | P           | Q          |
|----|--------------|-----------|----------------|-------------------|----------|--------------|-------------|---------|----------|--------|-------|------------|-------|-------------|--------|-------------|------------|
| 1  | FlightNumber | Date      | BoosterVersion | PayloadMass       | Orbit    | LaunchSite   | Outcome     | Flights | GridFins | Reused | Legs  | LandingPad | Block | ReusedCount | Serial | Longitude   | Latitude   |
| 2  | 1            | 6/4/2010  | Falcon 9       | 6123.547647058824 | LEO      | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B0003  | -80.577366  | 28.5618571 |
| 3  | 2            | 5/22/2012 | Falcon 9       |                   | 525 LEO  | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B0005  | -80.577366  | 28.5618571 |
| 4  | 3            | 3/1/2013  | Falcon 9       |                   | 677 ISS  | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B0007  | -80.577366  | 28.5618571 |
| 5  | 4            | 9/29/2013 | Falcon 9       |                   | 500 PO   | VAFB SLC 4E  | False Ocean | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B1003  | -120.610829 | 34.632093  |
| 6  | 5            | 12/3/2013 | Falcon 9       |                   | 3170 GTO | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B1004  | -80.577366  | 28.5618571 |
| 7  | 6            | 1/6/2014  | Falcon 9       |                   | 3325 GTO | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B1005  | -80.577366  | 28.5618571 |
| 8  | 7            | 4/18/2014 | Falcon 9       |                   | 2296 ISS | CCSFS SLC 40 | True Ocean  | 1       | FALSE    | FALSE  | TRUE  |            | 1     | 0           | B1006  | -80.577366  | 28.5618571 |
| 9  | 8            | 7/14/2014 | Falcon 9       |                   | 1316 LEO | CCSFS SLC 40 | True Ocean  | 1       | FALSE    | FALSE  | TRUE  |            | 1     | 0           | B1007  | -80.577366  | 28.5618571 |
| 10 | 9            | 8/5/2014  | Falcon 9       |                   | 4535 GTO | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B1008  | -80.577366  | 28.5618571 |
| 11 | 10           | 9/7/2014  | Falcon 9       |                   | 4428 GTO | CCSFS SLC 40 | None None   | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B1011  | -80.577366  | 28.5618571 |
| 12 | 11           | 6/24/2014 | Falcon 9       |                   | 2210 ISS | CCSFS SLC 40 | False Ocean | 1       | FALSE    | FALSE  | FALSE |            | 1     | 0           | B1012  | -80.577366  | 28.5618571 |

# Data Collection – Web Scrapping (overview)

- Additional datapoints were web-scraped from Wikipedia's "List of Falcon 9 and Falcon Heavy launches"
- URL:  
[https://en.wikipedia.org/w/index.php?title=List of Falcon 9 and Falcon Heavy launches&oldid=1027686922](https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922)

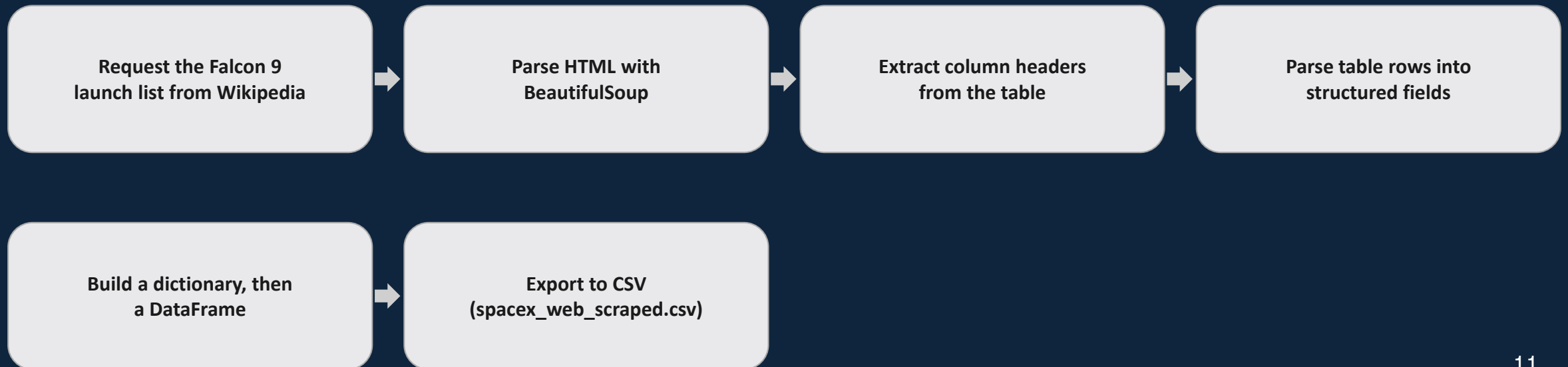
**2020** [\[ edit \]](#)

In late 2019, [Gwynne Shotwell](#) stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,<sup>[480]</sup> in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's *Long March* rocket family.<sup>[491]</sup>

| <a href="#">[hide]</a><br>Flight No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Date and time (UTC)                       | Version, Booster <sup>[2]</sup> | Launch site   | Payload <sup>[1]</sup>                                            | Payload mass                         | Orbit                       | Customer                    | Launch outcome | Booster landing      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------|---------------|-------------------------------------------------------------------|--------------------------------------|-----------------------------|-----------------------------|----------------|----------------------|
| 78                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7 January 2020, 02:19:21 <sup>[492]</sup> | F9 B5 Δ B1049.4                 | CCAFS, SLC-40 | Starlink 2 v1.0 (60 satellites)                                   | 15,600 kg (34,400 lb) <sup>[3]</sup> | LEO                         | SpaceX                      | Success        | Success (drone ship) |
| Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations. <sup>[493]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                                 |               |                                                                   |                                      |                             |                             |                |                      |
| 79                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 19 January 2020, 15:30 <sup>[494]</sup>   | F9 B5 Δ B1046.4                 | KSC, LC-39A   | Crew Dragon in-flight abort test <sup>[495]</sup> (Dragon C205.1) | 12,050 kg (26,570 lb)                | Sub-orbita <sup>[496]</sup> | NASA (CTS) <sup>[497]</sup> | Success        | No attempt           |
| An atmospheric test of the Dragon 2 abort system after <i>Max Q</i> . The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the <i>Crew Dragon Demo-1</i> capsule, <sup>[498]</sup> but that test article exploded during a ground test of SuperDraco engines on 20 April 2019. <sup>[416]</sup> The abort test used the capsule originally intended for the first crewed flight. <sup>[499]</sup> As expected, the booster was destroyed by aerodynamic forces after the capsule aborted. <sup>[500]</sup> First flight of a Falcon 9 with only one functional stage — the second stage had a <i>mass simulator</i> in place of its engine. |                                           |                                 |               |                                                                   |                                      |                             |                             |                |                      |
| 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 29 January 2020, 14:07 <sup>[501]</sup>   | F9 B5 Δ B1051.3                 | CCAFS, SLC-40 | Starlink 3 v1.0 (60 satellites)                                   | 15,600 kg (34,400 lb) <sup>[3]</sup> | LEO                         | SpaceX                      | Success        | Success (drone ship) |
| Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean. <sup>[502]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                                 |               |                                                                   |                                      |                             |                             |                |                      |
| 81                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 17 February 2020, 15:05 <sup>[503]</sup>  | F9 B5 Δ B1056.4                 | CCAFS, SLC-40 | Starlink 4 v1.0 (60 satellites)                                   | 15,600 kg (34,400 lb) <sup>[3]</sup> | LEO                         | SpaceX                      | Success        | Failure (drone ship) |
| Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship <sup>[504]</sup> due to incorrect wind data. <sup>[505]</sup> This was the first time a flight proven booster failed to land.                                                                                                                                                                                                                                                                                                                                                                                             |                                           |                                 |               |                                                                   |                                      |                             |                             |                |                      |
| 82                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7 March 2020, 04:50 <sup>[506]</sup>      | F9 B5 Δ B1059.2                 | CCAFS, SLC-40 | SpaceX CRS-20 (Dragon C112.3 Δ)                                   | 1,977 kg (4,359 lb) <sup>[507]</sup> | LEO (ISS)                   | NASA (CRS)                  | Success        | Success (ground pad) |
| Last launch of phase 1 of the CRS contract. Carries <i>Berlino</i> , an ESA platform for hosting external payloads onto ISS. <sup>[508]</sup> Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part. <sup>[509]</sup> It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo <i>Dragon</i> spacecraft.                                                                                                                                                                                                                                                                                                     |                                           |                                 |               |                                                                   |                                      |                             |                             |                |                      |
| 83                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 18 March 2020, 12:16 <sup>[510]</sup>     | F9 B5 Δ B1048.5                 | KSC, LC-39A   | Starlink 5 v1.0 (60 satellites)                                   | 15,600 kg (34,400 lb) <sup>[3]</sup> | LEO                         | SpaceX                      | Success        | Failure (drone ship) |
| Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019). <sup>[511]</sup> Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a <i>Merlin 1D</i> variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit. <sup>[512]</sup> This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor. <sup>[513]</sup>                                                                                                                                                                                   |                                           |                                 |               |                                                                   |                                      |                             |                             |                |                      |
| 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 22 April 2020, 19:30 <sup>[514]</sup>     | F9 B5 Δ B1051.4                 | KSC, LC-39A   | Starlink 6 v1.0 (60 satellites)                                   | 15,600 kg (34,400 lb) <sup>[3]</sup> | LEO                         | SpaceX                      | Success        | Success (drone ship) |

# Data Collection – Web Scraping (methodology)

Flowchart describes the processing  
pipeline for webscraping and  
preparation of Wikipedia data



# Data Collection – Web Scrapping (results)

Data scraped from Wikipedia was stored in spacex\_web\_scraped.csv. A sample is below. 121 records in total were retrieved from Wikipedia for the period in question.

|    | A          | B              | C                                    | D            | E           | F        | G              | H                | I               | J         | K     |
|----|------------|----------------|--------------------------------------|--------------|-------------|----------|----------------|------------------|-----------------|-----------|-------|
| 1  | Flight No. | Launch site    | Payload                              | Payload mass | Orbit       | Customer | Launch outcome | Version Booster  | Booster landing | Date      | Time  |
| 2  | 1          | CCAFS          | Dragon Spacecraft Qualification Unit |              | 0 LEO       | SpaceX   | Success        | F9 v1.07B0003.18 | Failure         | 4-Jun-10  | 18:45 |
| 3  | 2          | CCAFS          | Dragon                               |              | 0 LEO       | NASA     | Success        | F9 v1.07B0004.18 | Failure         | 8-Dec-10  | 15:43 |
| 4  | 3          | CCAFS          | Dragon                               | 525 kg       | LEO         | NASA     | Success        | F9 v1.07B0005.18 | NoÂ attempt     | 22-May-12 | 7:44  |
| 5  | 4          | CCAFS          | SpaceX CRS-1                         | 4,700 kg     | LEO         | NASA     | Success        | F9 v1.07B0006.18 | No attempt      | 8-Oct-12  | 0:35  |
| 6  | 5          | CCAFS          | SpaceX CRS-2                         | 4,877 kg     | LEO         | NASA     | Success        | F9 v1.07B0007.18 | NoÂ attempt     | 1-Mar-13  | 15:10 |
| 7  | 6          | VAFB           | CASSIOPE                             | 500 kg       | Polar orbit | MDA      | Success        | F9 v1.17B10038   | Uncontrolled    | 29-Sep-13 | 16:00 |
| 8  | 7          | CCAFS          | SES-8                                | 3,170 kg     | GTO         | SES      | Success        | F9 v1.1          | NoÂ attempt     | 3-Dec-13  | 22:41 |
| 9  | 8          | CCAFS          | Thaicom 6                            | 3,325 kg     | GTO         | Thaicom  | Success        | F9 v1.1          | NoÂ attempt     | 6-Jan-14  | 22:06 |
| 10 | 9          | Cape Canaveral | SpaceX CRS-3                         | 2,296 kg     | LEO         | NASA     | Success        | F9 v1.1          | Controlled      | 18-Apr-14 | 19:25 |
| 11 | 10         | Cape Canaveral | Orbcomm-OG2                          | 1,316 kg     | LEO         | Orbcomm  | Success        | F9 v1.1          | Controlled      | 14-Jul-14 | 15:15 |
| 12 | 11         | Cape Canaveral | AsiaSat 8                            | 4,535 kg     | GTO         | AsiaSat  | Success        | F9 v1.1          | NoÂ attempt     | 5-Aug-14  | 8:00  |
| 13 | 12         | Cape Canaveral | AsiaSat 6                            | 4,428 kg     | GTO         | AsiaSat  | Success        | F9 v1.1          | NoÂ attempt     | 7-Sep-14  | 5:00  |



# Data Wrangling (methodology)

Landing outcomes in the raw data include several success/failure variants — True/False Ocean, RTLS (return to launch site), and ASDS (drone ship). These were consolidated into a single binary training label: Class = 1 for any successful landing, Class = 0 for any failure or no attempt.

- Calculated launch counts per site
- Calculated occurrence of each orbit type
- Cross-tabulated mission outcome by orbit type
- Derived the binary landing Class label from the Outcome column
- Exported the cleaned dataset to CSV (dataset\_part\_2.csv)

A decorative header consisting of a row of hexagons in various shades of blue, orange, and white. Below this row, a second row of hexagons is partially visible, with some outlined in white and others filled with solid colors.

# Exploratory Data Analysis (EDA)

- Graphical Analysis of Prepared Data
- Database Exploration



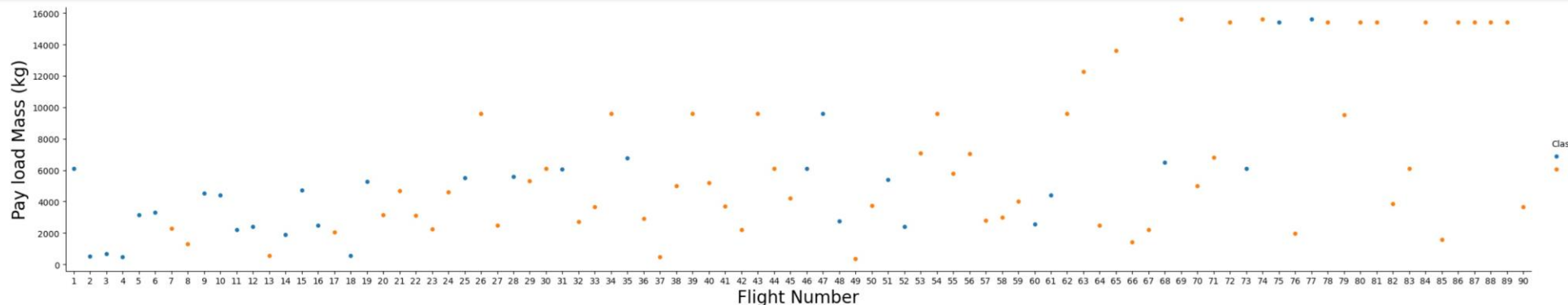
# EDA with Data Visualization

Scatter plots reveal relationships between numeric variables that may be useful features for the classification model. Bar charts compare success rate across discrete categories (launch site, orbit). The line chart shows how the success rate trends over time.

- Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site
- Success Rate vs. Orbit Type, Flight Number vs. Orbit Type, Payload Mass vs. Orbit Type
- Success Rate Yearly Trend

# EDA - Visual Payload vs Flight #

- Newer flights (higher flight number) tend to carry heavier payloads
- Later flights skew toward successful landings (orange) regardless of payload size
- Suggests SpaceX's landing technique matured over time, independent of payload mass

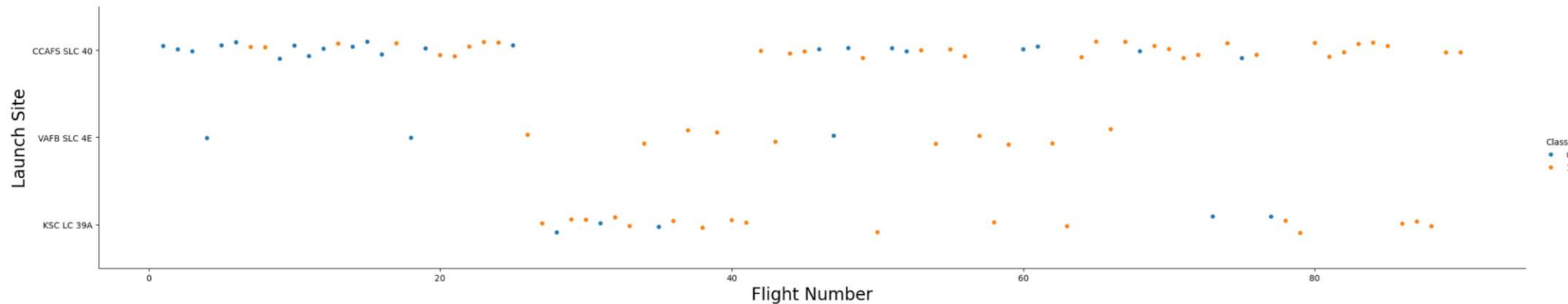




# EDA - Visual Launch Site vs Flight #

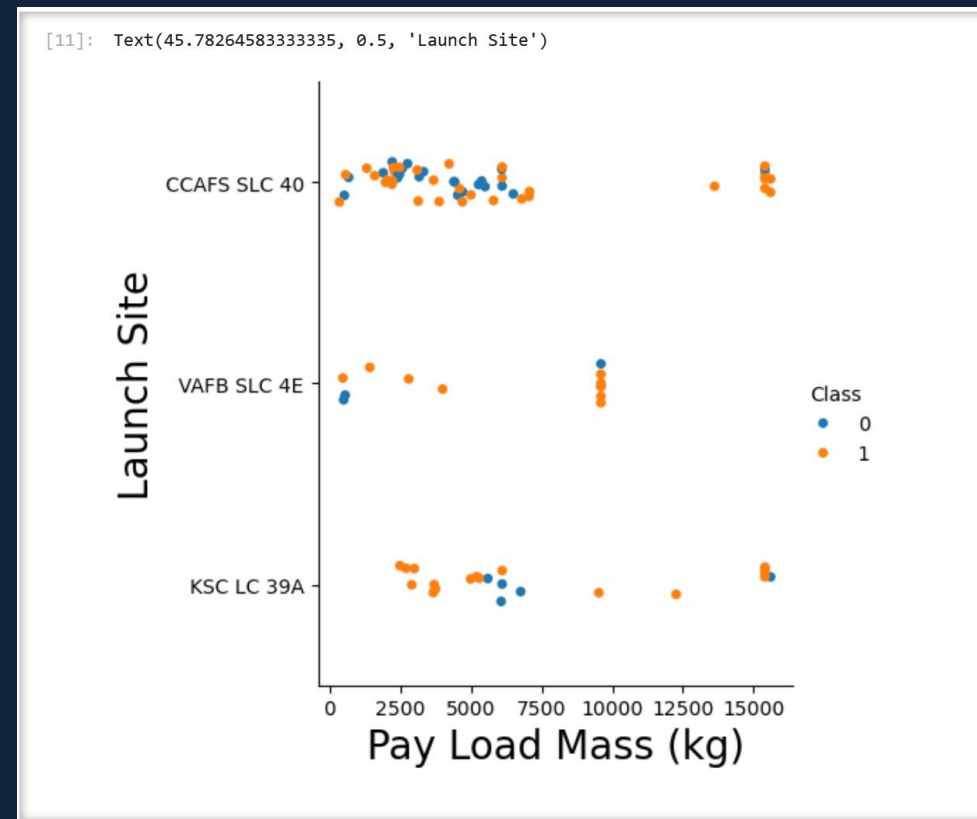
- The earliest flights mostly failed; the most recent flights mostly succeeded
- CCAFS SLC-40 hosts roughly half of all launches
- VAFB SLC-4E and KSC LC-39A show higher success rates
- Each successive launch is more likely to succeed than the last

[10]: Text(32.35652916666665, 0.5, 'Launch Site')



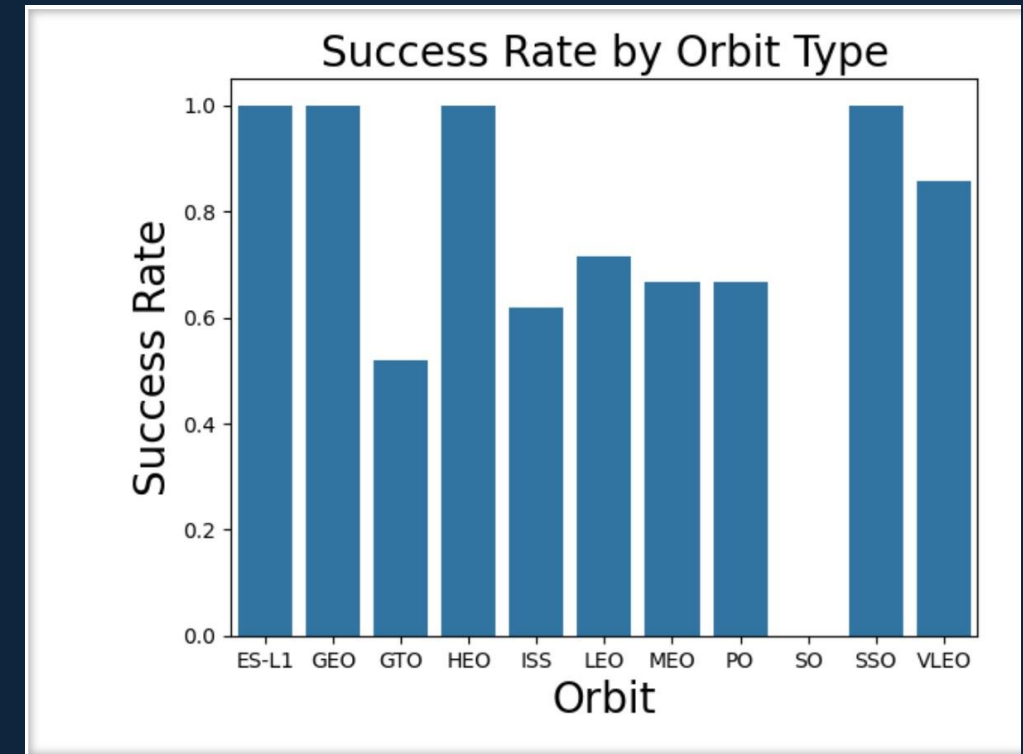
# EDA - Visual Launch Site vs Payload

- At every site, heavier payloads correlate with a higher success rate
- Most launches carrying over 7,000 kg succeeded
- KSC LC-39A also achieves a 100% success rate for payloads under 5,500 kg



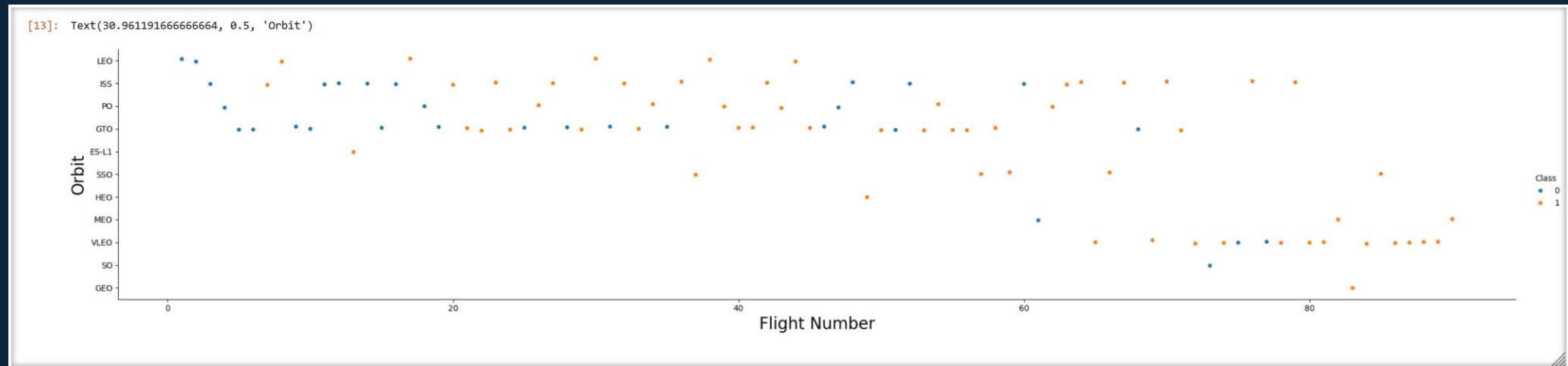
# EDA - Visual Success Rate vs Orbit

- 100% success: ES-L1, GEO, HEO, SSO
- 0% success: SO
- 50-85% success: GTO, ISS, LEO, MEO, PO
- Orbit type is a strong candidate predictive feature



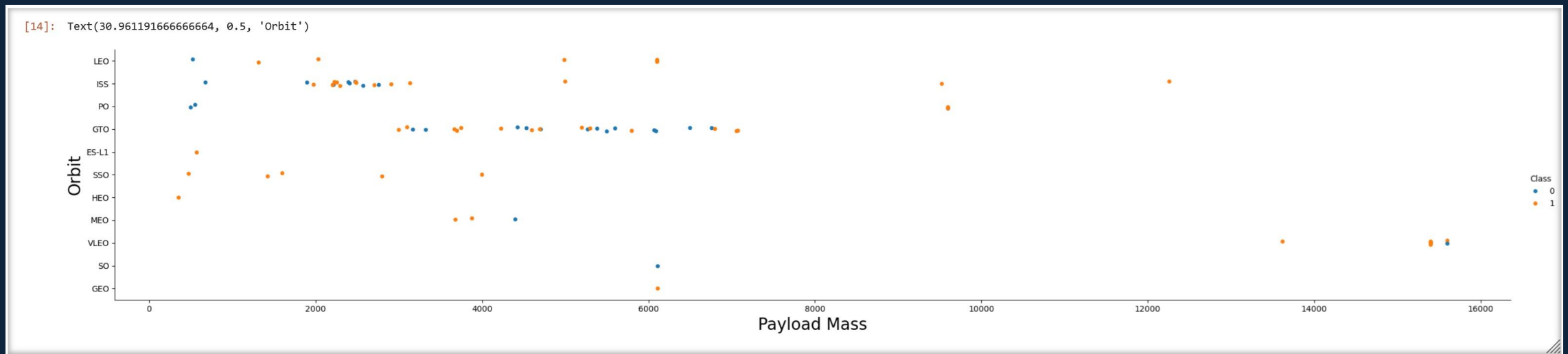
# EDA - Visual Orbit vs Flight #

- In LEO, success appears related to flight number — later LEO flights succeed more often
- In GTO, flight number shows little to no relationship with success
- Indicates the learning curve effect is orbit-dependent



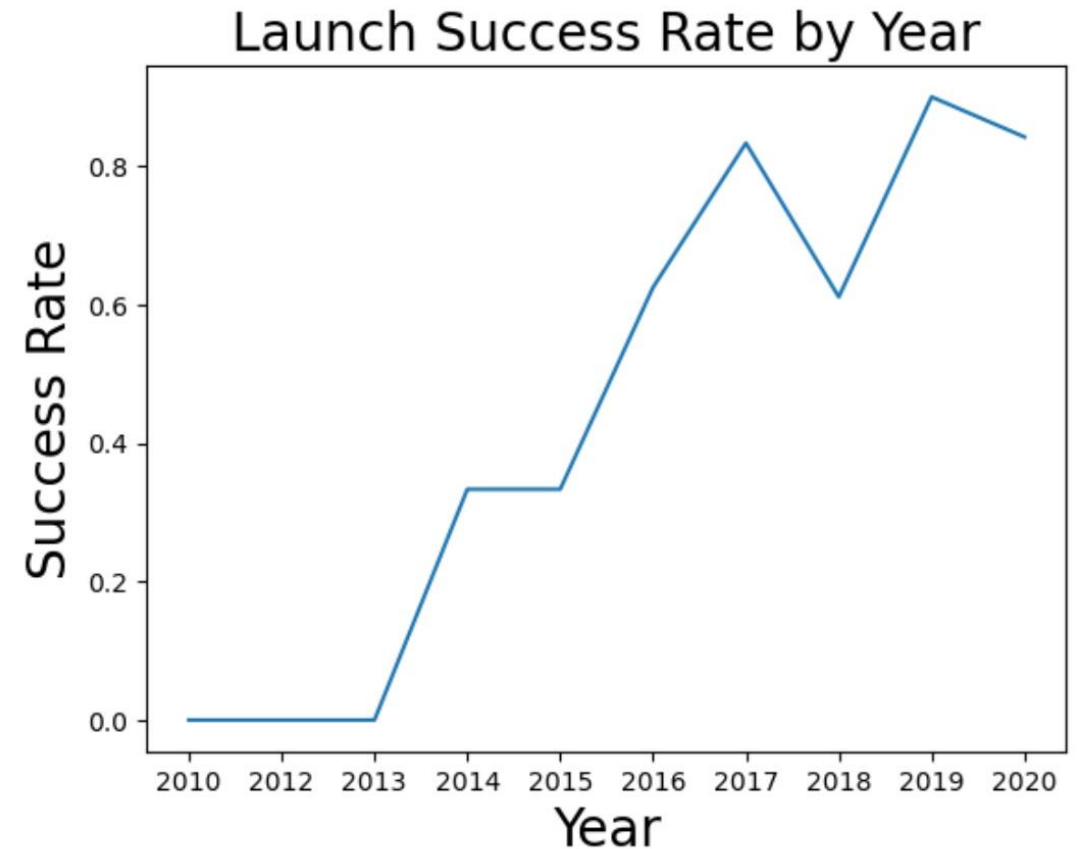
# EDA - Visual Orbit vs Payload

- Heavy payloads appear to hurt success in GTO orbit
- Heavy payloads appear to help success in LEO/ISS (Polar LEO) orbit
- Orbit and payload mass likely interact rather than acting independently



# EDA - Visual Success Rate vs Year

- Success rate was 0% from 2010-2013
- Steady increase from 2013 through 2020
- Reflects SpaceX's iterative improvement of landing technique over time





# EDA with SQL (overview)

Data collected and prepared from the existing data sources were loaded into a database for continued analysis.



# EDA with SQL (queries)

Queries were written in SQL to retrieve answers to each of the questions listed:

- Unique launch site names
- Records where the launch site begins with 'CCA'
- Total payload mass carried for NASA (CRS) missions
- Average payload mass for booster version F9 v1.1
- Date of the first successful ground-pad landing
- Boosters with a successful drone-ship landing carrying 4,000-6,000 kg payloads
- Total count of successful vs. failed mission outcomes
- Booster version(s) that carried the maximum payload mass
- 2015 drone-ship landing failures, by booster and launch site
- Ranked count of landing outcomes between 2010-06-04 and 2017-03-20



# EDA with SQL (findings)

Executing the SQL queries against the database created from our initial data-load yielded the following results:

| Query                                          | Result                                                              |
|------------------------------------------------|---------------------------------------------------------------------|
| Unique launch sites                            | CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E                  |
| Total payload mass for NASA (CRS)              | 48,213 kg                                                           |
| Average payload mass, F9 v1.1                  | 2,534.7 kg                                                          |
| First successful ground-pad landing            | 2015-12-22                                                          |
| Successful drone-ship landings, 4,000-6,000 kg | F9 FT B1022, B1026, B1021.2, B1031.2                                |
| Total mission outcomes                         | Success: 98   Success (payload unclear): 1   Failure (in flight): 1 |
| Booster carrying max payload mass              | F9 B5 (multiple serials, e.g. B1048-B1060 series)                   |
| 2015 drone-ship landing failures               | F9 v1.1 B1012 (Jan), F9 v1.1 B1015 (Apr) — both at CCAFS LC-40      |
| Top landing outcomes, 2010-06-04 to 2017-03-20 | No attempt (10), Success drone ship (6), Failure drone ship (5)     |



# Interactive Visual Analytics

To obtain further insight, two interactive visual tools were developed to facilitate deeper exploration of several factors.

- GIS – A GIS interface was created using Folium Map to explore launch sites and their success/failure rates and to investigate how their location may or may not impact mission success and/or safety concerns.
- Dashboard – A Dashboard was created using Plotly Dash to compare launch site success rates and to examine patterns pertaining to payload mass on mission successes.



# Interactive Visual Analytics GIS

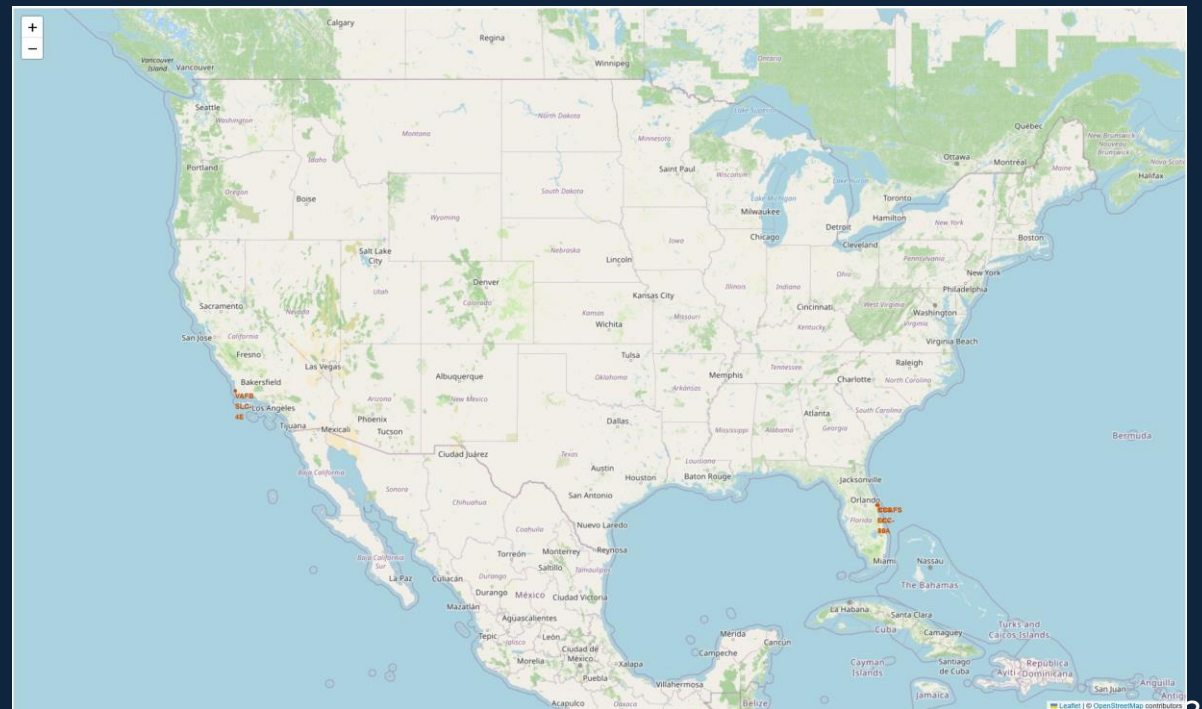
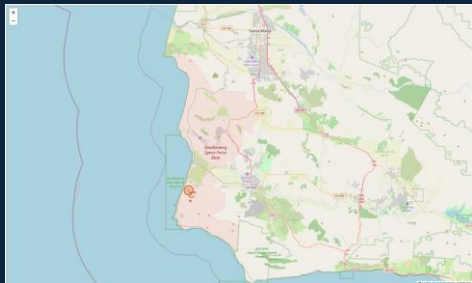
- Folium Map
- Added a marker for NASA Johnson Space Center as a reference start location
- Added a marker (with circle + label) for each of the 4 launch sites, using lat/long
- Color-coded individual launch outcomes per site using MarkerCluster — green = success, red = failure
- Drew distance lines from KSC LC-39A to the nearest railway, highway, coastline, and city

GitHub: [03-1-lab\\_jupyter\\_launch\\_site\\_location.ipynb](#)

# GIS

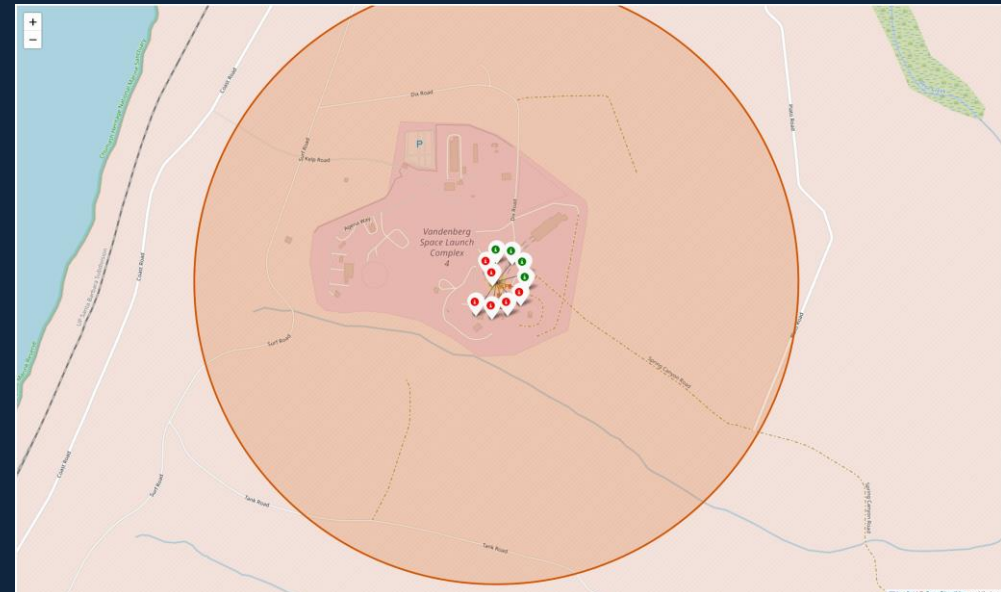
## US Launch Sites

- All 4 launch sites are plotted with markers and labels
- All sites sit close to the Equator, which gives a rotational speed boost to orbit
- All sites sit close to the coast, reducing risk to populated areas from launch failures



# GIS VAFB

- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate



# GIS

## KSC LC39A

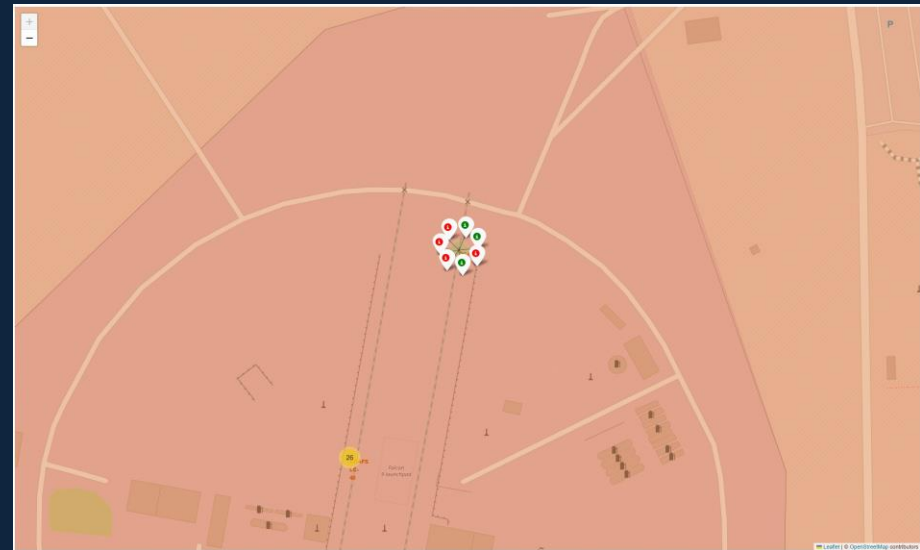
- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate



# GIS

## CCAFS SLC40 (a)

- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate

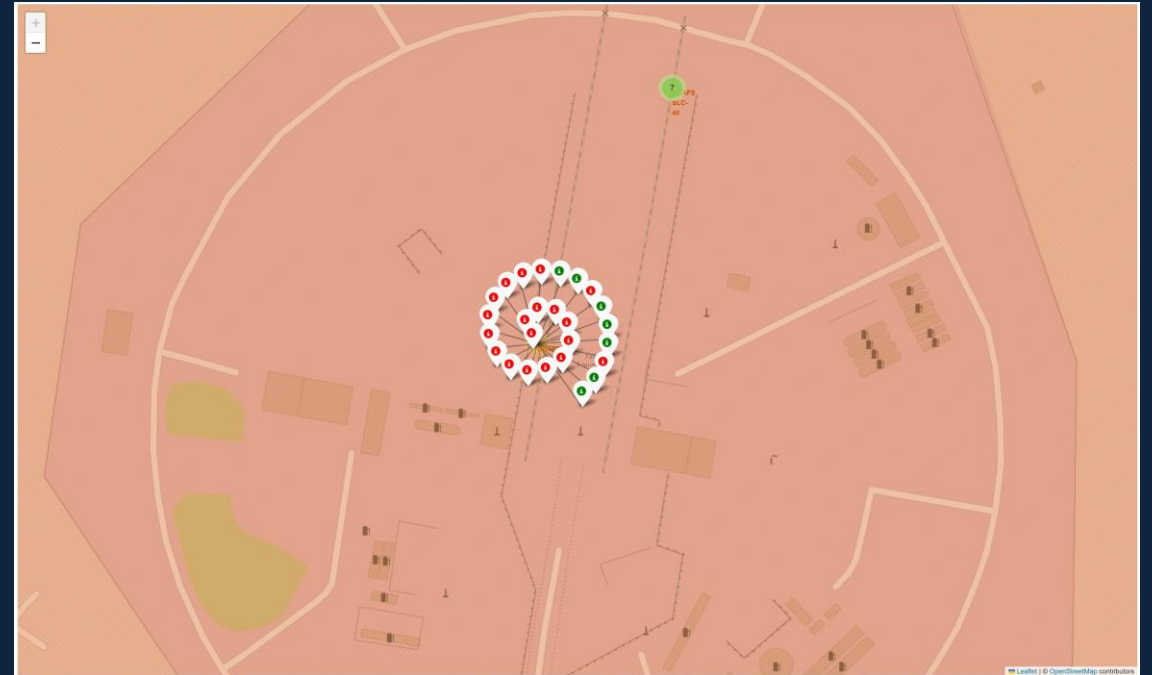




# GIS

## CCAFS SLC40 (b)

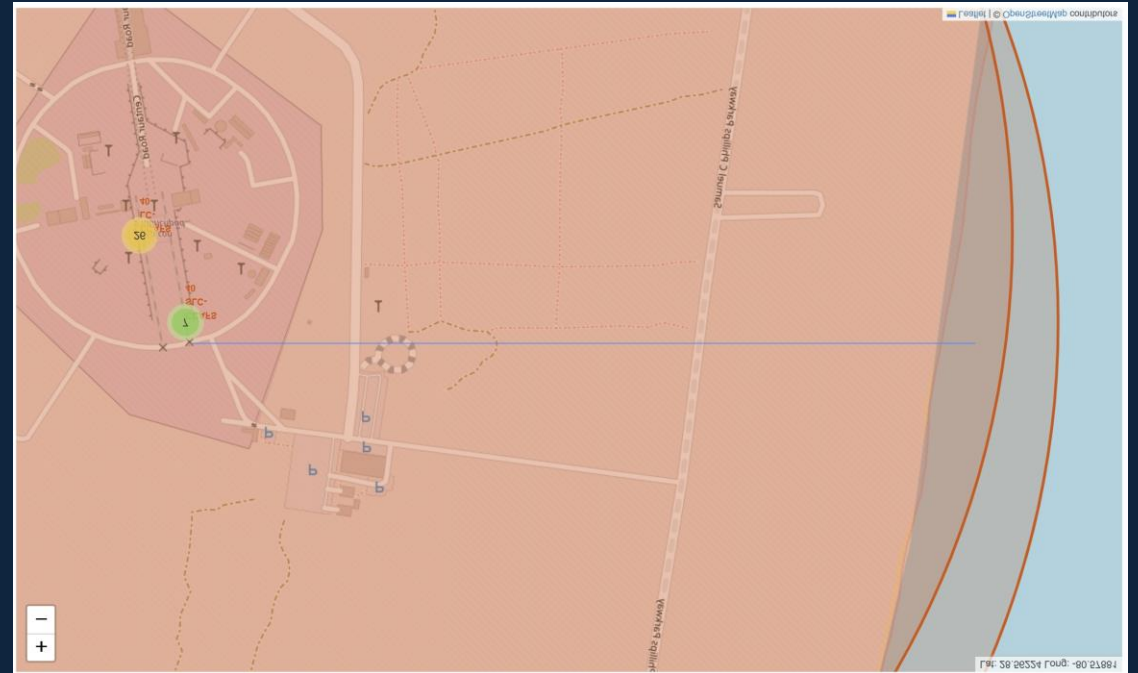
- Green markers = successful landing, red = failed landing
- Marker clustering makes per-site success patterns easy to scan visually
- KSC LC-39A clearly shows a majority-green cluster, consistent with its high success rate





# GIS Proximity Review

- Distance lines drawn from the launch site to nearby railway, highway, coastline, and city
- Confirms launch sites are sited away from dense population centers, favoring closeness to the coast
- A failed rocket at high speed could travel the railway/highway distance within seconds
- Supports the safety rationale behind launch site placement

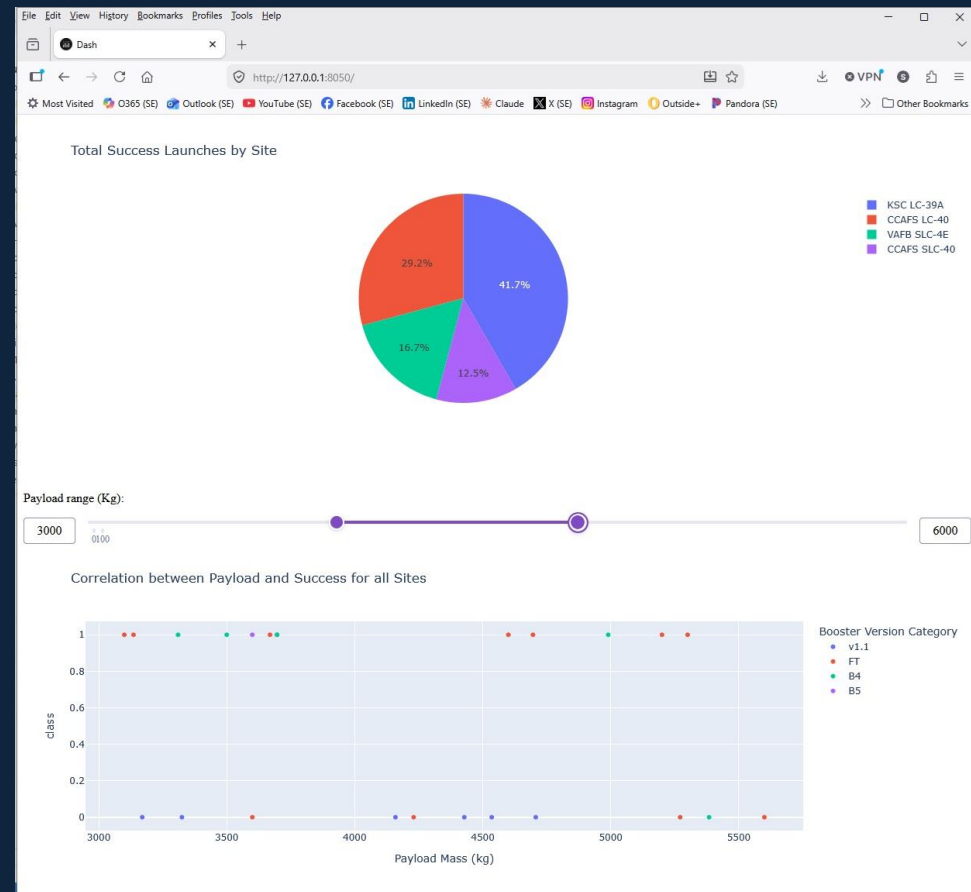


# Interactive Visual Analytics Dashboard

An interactive dashboard was developed to provide further visual analysis.

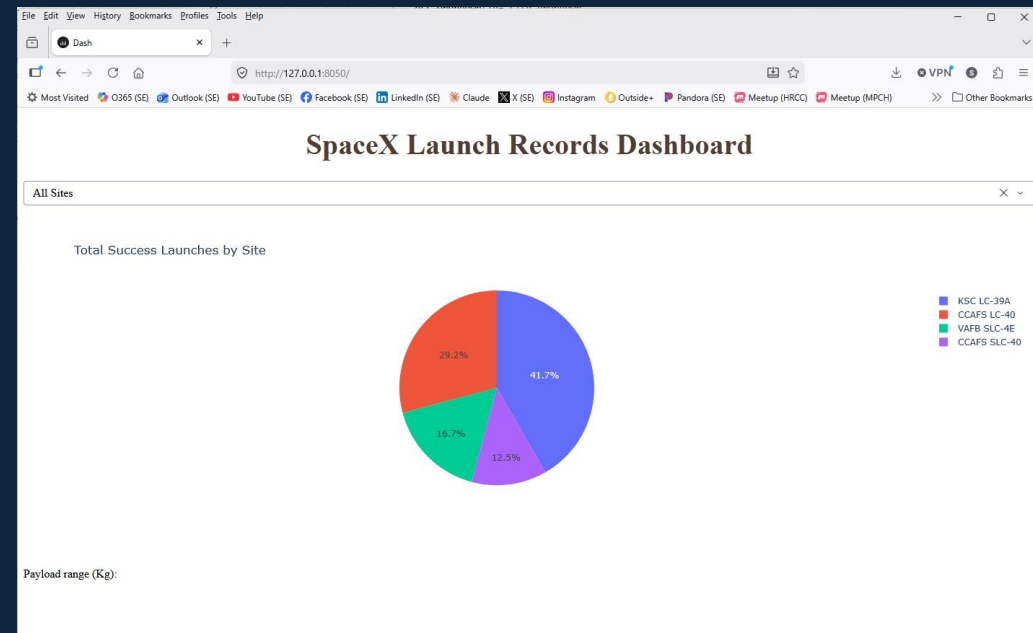
- Plotly Dash
- Dropdown lets the user pick a single launch site or All Sites
- Payload range can be adjusted by means of a slider control
- Dropdown list to select a launch site (or All Sites)
- Pie chart of successful launches — all sites, or success/failure split for one selected site
- Range slider to filter by payload mass
- Scatter chart of payload mass vs. launch outcome, colored by booster version

GitHub: [03-2-spacex\\_dash\\_app.py](#)



# Dashboard Piechart (All)

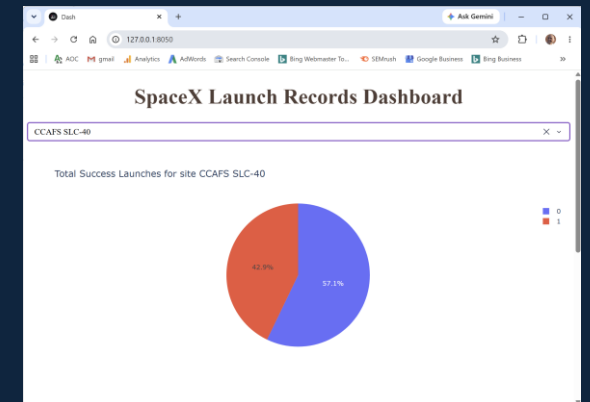
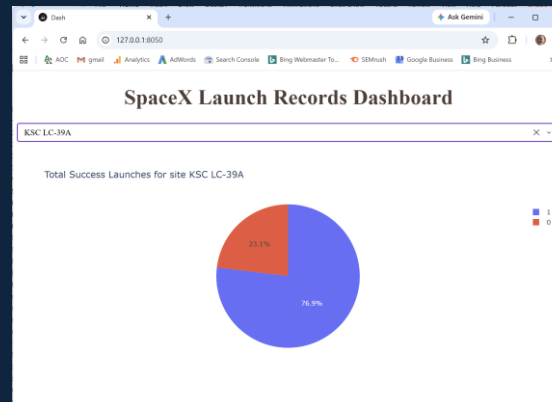
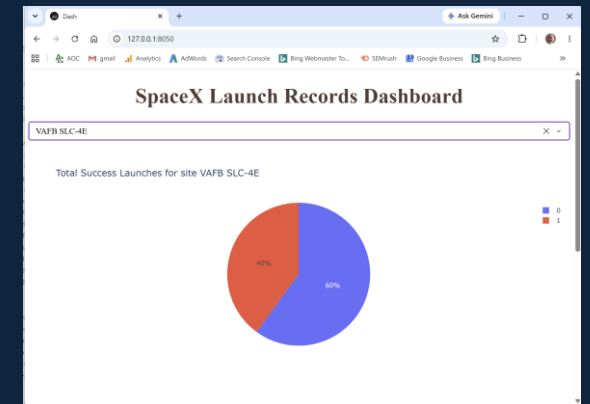
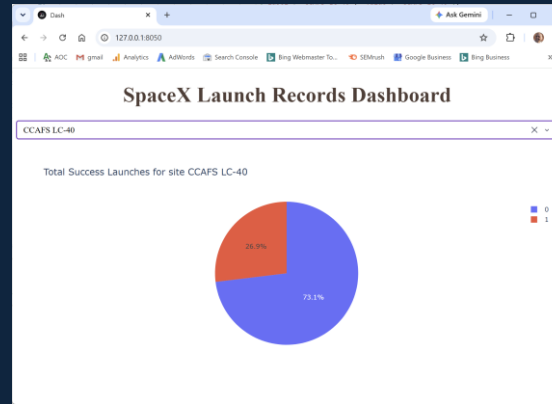
- All Sites
- KSC LC-39A accounts for the largest share (41.2%) of successful launches
- CCAFS SLC-40 follows at 23%, then VAFB SLC-4E at 21.4%, then CCAFS LC-40 at 14.4%



# Dashboard Piechart (Select)

Using the Launch Site selector, success/failure ratios could be examined for each site

- CCAFS LC-40: 73.1% success / 26.9% fail
- VAFB SLC-4E: 60% success / 40% fail
- KSC LC-39A: 76.9% success / 23.1% fail
- CCAFS SLC-40: 57.1% success / 42.9% fail



# Dashboard Payload vs Outcome

- Scatter Plot
- Full Range

- Payload range slider filters the scatter plot live
- Points colored by booster version category (v1.0, v1.1, FT, B4, B5)
- Payloads roughly between 2,000 and 5,500 kg show the strongest success rate



# Dashboard Payload vs Outcome

- Scatter Plot
- Partial Range



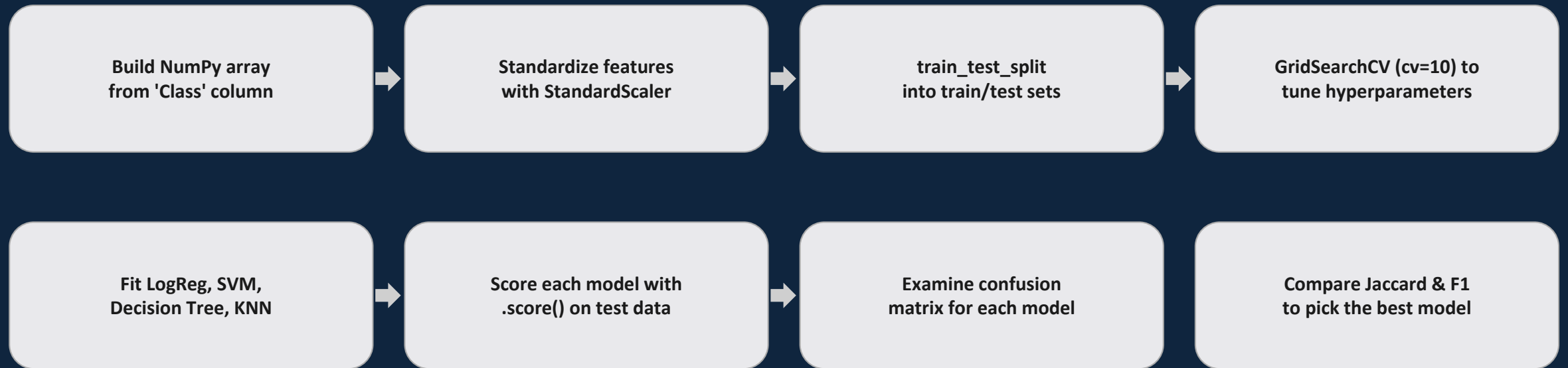
A decorative header consisting of a grid of hexagons in various shades of blue, orange, and white. Some hexagons are solid, while others are outlined.

# Predictive Analysis

- Tuned w/ GridSearchCV (cv=10)
- Four Models
  - Logistic Regression
  - SVM
  - Decision Tree
  - KNN



# Predictive Analysis - Methodology

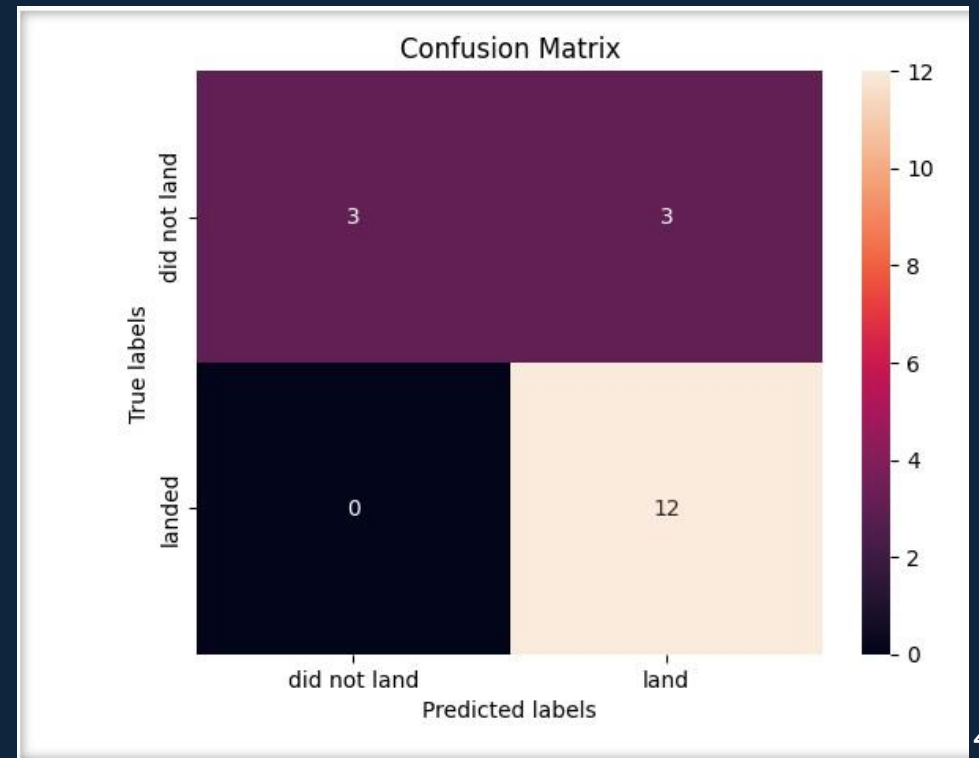




# Predictive Analysis

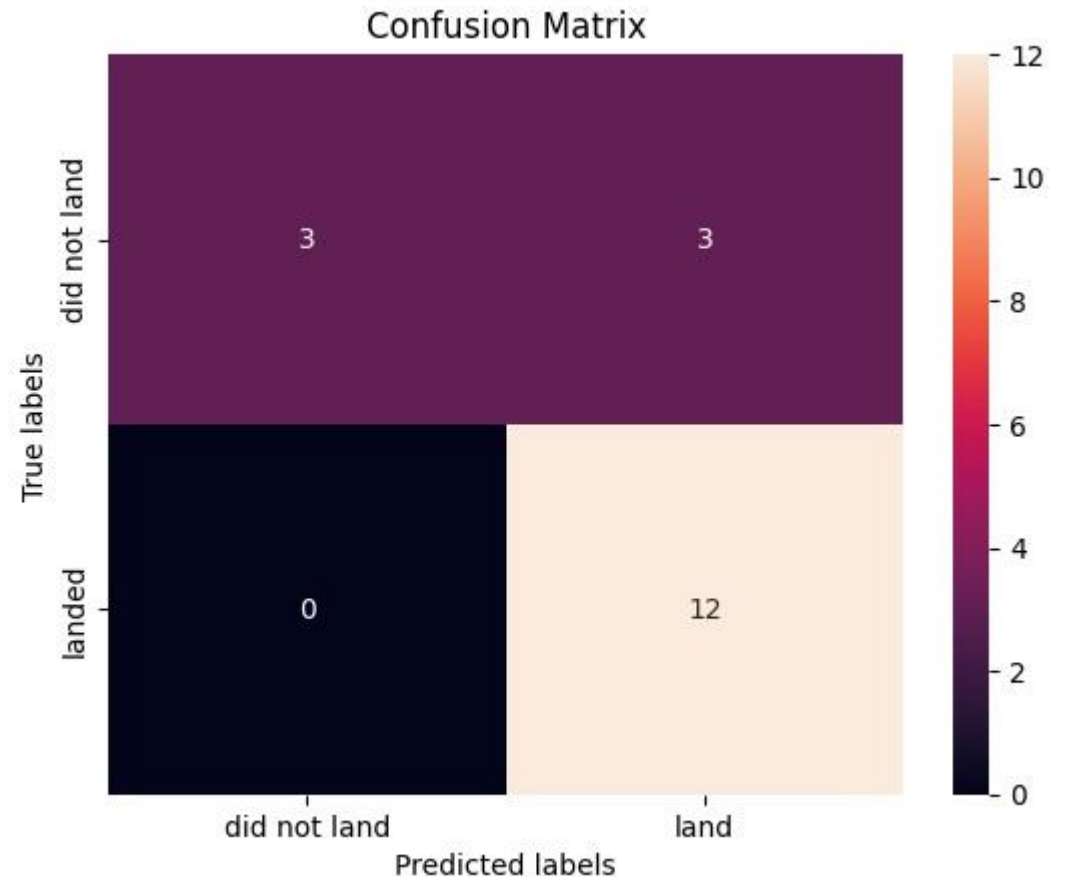
## Logistic Regression

- Model: Logistic Regression
- Test Set Accuracy: 0.8333
- Best CV Accuracy: 0.846429
- Model correctly identifies all 12 successful landings (no false negatives)
- 3 of 6 failed landings are misclassified as successes (false positives)
- False positives are the dominant error mode for this model



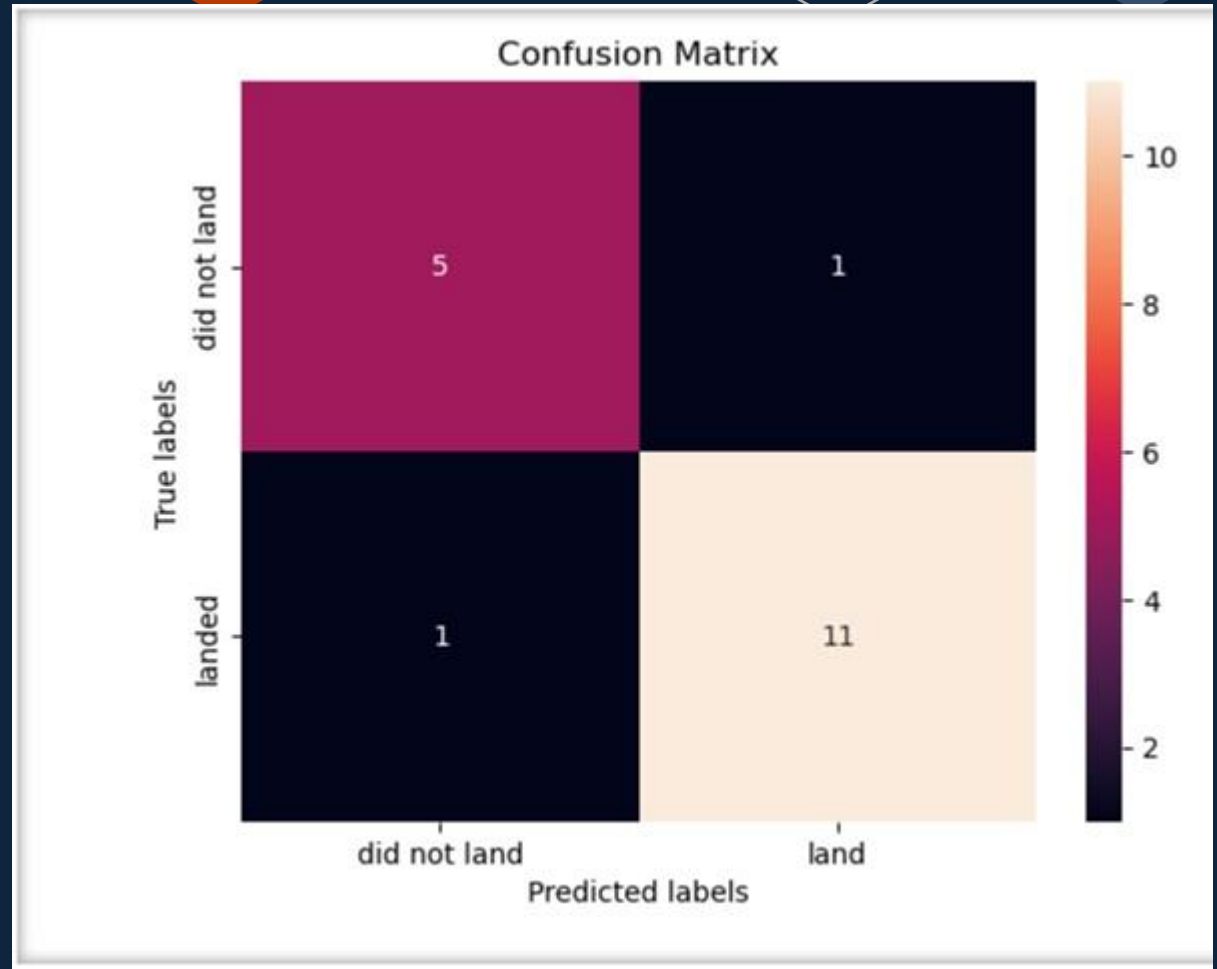
# Predictive Analysis SVM

- Model: SVM
- Test Set Accuracy: 0.8333
- Best CV Accuracy: 0.848214
- Model correctly identifies all 12 successful landings (no false negatives)
- 3 of 6 failed landings are misclassified as successes (false positives)
- False positives are the dominant error mode for this model



# Predictive Analysis Decision Tree

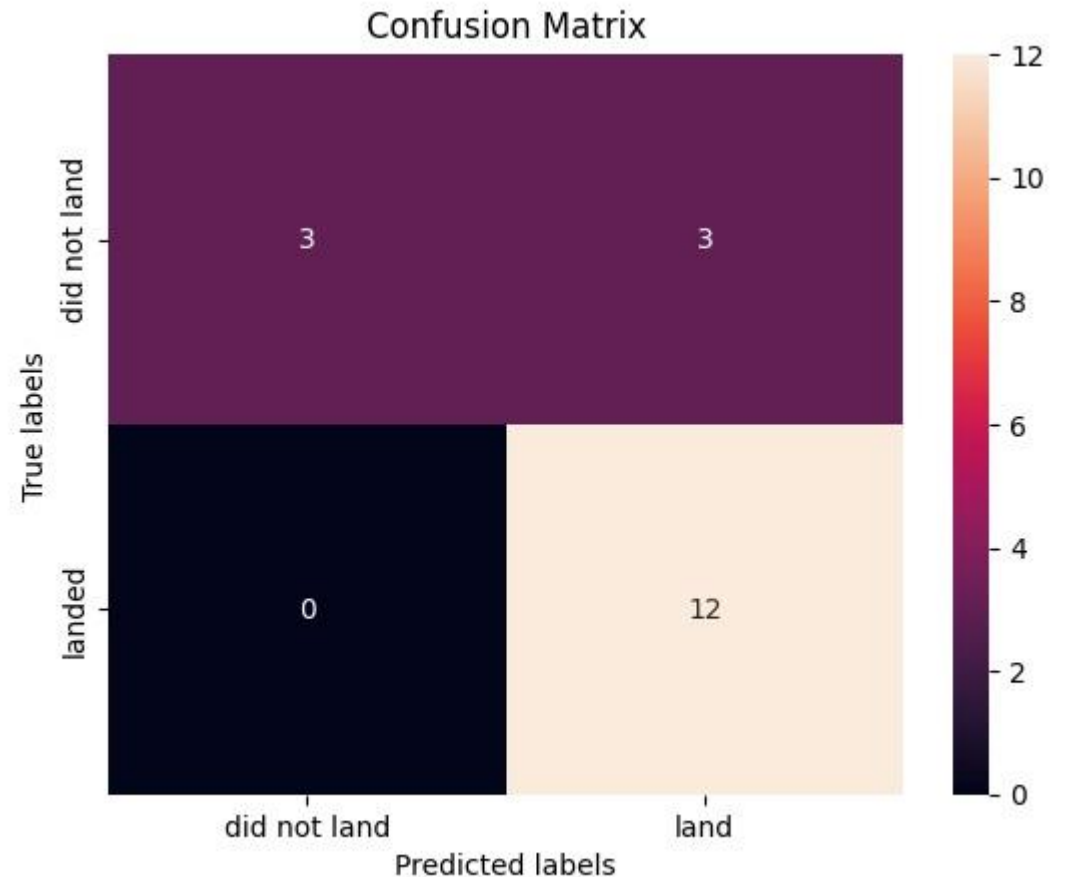
- Model: Decision Tree
- Test Set Accuracy: 0.8888
- Best CV Accuracy: 0.889286
- Model correctly identifies 11 successful landings
- Model correctly identifies 5 failed landings
- 1 of 6 failed landings are misclassified as successes (false positives)
- 1 of 6 successful landings are misclassified as failures (false negatives)



# Predictive Analysis

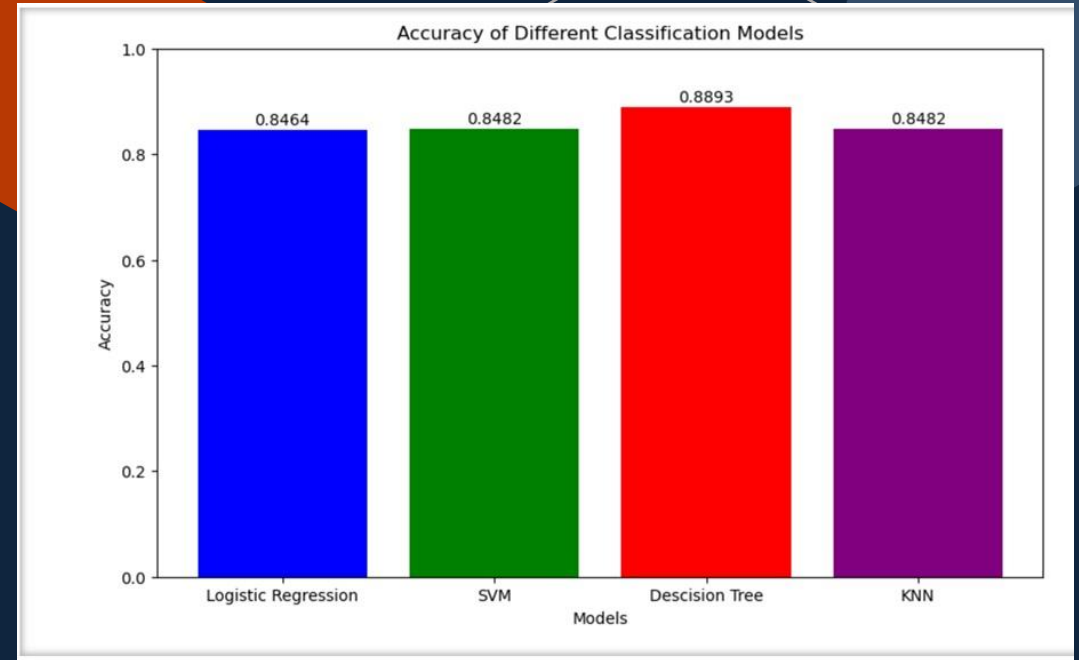
## KNN

- Model: KNN
- Test Set Accuracy: 0.8333
- Best CV Accuracy: 0.848214
- Model correctly identifies all 12 successful landings (no false negatives)
- 3 of 6 failed landings are misclassified as successes (false positives)
- False positives are the dominant error mode for this model



# Predictive Analysis Best Model

- Best Model: Decision Tree Classifier
- Decision Tree achieves the highest accuracy (0.8750) of the four tuned models
- Logistic Regression, SVM, and KNN tie at 0.8333 on this comparison
- Decision Tree is selected as the best model for predicting first-stage landing



| Model               | Best CV Accuracy (GridSearchCV) | Test Set Accuracy |
|---------------------|---------------------------------|-------------------|
| Logistic Regression | 0.846429                        | 0.8333            |
| SVM                 | 0.848214                        | 0.8333            |
| Decision Tree       | 0.889286                        | 0.8888            |
| KNN                 | 0.848214                        | 0.8333            |



# Discussion

- Landing success is driven by a combination of factors, not any single variable: launch site, payload mass, and orbit type all interact
- The steady year-over-year improvement in success rate reflects SpaceX's iterative engineering process rather than a one-time fix
- Mid-range payloads (roughly 2,000-5,500 kg) and orbits like GEO/HEO/SSO are the most reliable bets for a successful landing
- All four classifiers perform similarly on raw test accuracy because of the small dataset (90 launches); cross-validation is essential to distinguish them
- False positives (predicting a landing when none occurred) are the main error mode — a meaningful limitation if this model were used to estimate launch cost in advance



# Conclusions

- The Decision Tree model is the best-performing algorithm for predicting Falcon 9 first-stage landing outcomes (87.5% cross-validated accuracy)
- Launch site, payload mass, and orbit type all materially affect landing success and should be treated as core predictive features
- Landing success has trended upward every year since 2013, reflecting SpaceX's maturing recovery process
- KSC LC-39A is the strongest-performing launch site, with a 76.9% success rate
- Orbits ES-L1, GEO, HEO, and SSO have a 100% observed success rate in this dataset
- Because the first stage is reusable when it lands successfully, this kind of model gives a data-driven way to estimate the cost of a future SpaceX launch

A decorative pattern of hexagons in various shades of blue, orange, and white, arranged in a honeycomb-like structure on the left side of the slide.

# Thank you

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<https://github.com/StephenElmer>